

# Retaining Software Engineers and Engineering Managers

Predictors of attrition, and how to get ahead of them in tech organizations.



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MAR 29, 2022 • PAID



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## Q: As an engineering manager, what are things in my control that I can do to retain software engineers?

This question could not be more timely. We are still in the middle of one of the hottest job markets I've seen, and at the beginning of what might be '[The Great Reshuffle](#)': more people in tech switching jobs more rapidly, with many entering the job market for the first time since the pandemic started.

In this issue we cover:

- Four predictors of attrition and getting ahead of them.
- Early-career engineers: advice on how to retain them.
- Retaining software engineers.
- Engineering managers: retention approaches.

## Four predictors of attrition

If you ask a hundred people why they quit their job, you'd get close to a hundred different answers. Every situation is unique, and so is the reason for leaving.

However, after talking with dozens of engineers and managers who changed jobs and observing even more people do so, here are the four root causes of why someone leaves:

- Money & career
- Environment
- Caring
- Growth

## Retaining software engineers and engineering managers



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**As a manager, get a sense of these four areas' priority and importance for each of your team members.** Every individual will assign different importance to them and the order of priorities might change as their situation changes. For example, one engineer on my team had career progression as their top priority for a few years. However, as

their family situation changed, they decided for career growth to take a back seat, in favor of doing shorter hours and spending more time with their family. A healthy and supportive professional environment became their biggest priority.

I've observed that if someone is happy in all four areas, they are unlikely to consider new positions. However, if someone feels unhappy in two or more areas, they're likely on the verge of actively looking for new opportunities.

## Money & career

A colleague's unhappiness with a job's remuneration or career prospects can easily turn into attrition risk for their manager. There are, of course, situations when people don't mind being paid below the market, or not having career progression lined up. However, I'd suggest these are the exceptions.

The most common pain points of this area:

**Compensation.** If people are paid below the market, expect attrition. Most people have a hard time saying "no" to offers that are 30% or more, above what they make. Attrition due to compensation falling behind the market is more common across the industry than ever before. Over the past 18 months, the market [has moved up](#) about 20-40% for senior engineers, 15-30% for mid-level engineers and close to nothing for entry-level engineers. I cover this change in [The perfect storm causing an insane tech hiring market](#), and several issues of [The Scoop](#). In the November issue of [More follow-up on the tech hiring market](#) I write:

"Advice for hiring managers: rely less on your usual compensation data sources, and collect your own market data. Consider a differentiated compensation adjustment approach based on seniority and attrition risk."

**Rejecting promotions.** People up for promotion are usually some of the top performers on your team. However, most promotion processes reject a good percentage of candidates.

What happens to people who want promotion, but get rejected? In today's hot tech market, they instantly become a flight risk. I've observed this happen first-hand, and

have talked to half a dozen engineers who started job hunting after they did not get the promotion their manager supported.

Don't put people up for promotion if the outcome is uncertain. Expect everyone rejected for a promotion to be an attrition risk while this hot market persists.

**Predictable and fair performance reviews.** Performance reviews that surprise people with bad news are a common reason why they decide to leave. As I write in [my article on performance reviews](#):

"Often, these bad performance reviews marked my "shields down" moment. The point I lost trust in my manager and decided that I need to change teams or companies."

Avoid surprises in performance reviews, it will reduce the risk of people leaving because of unexpected bad news.

**Reducing compensation.** With remote work gaining ground, several companies are allowing remote work. However, even more are implementing location-based compensation which means they reduce pay for people who move to places with lower costs of living. As I report in [issues of the Scoop](#), both Google and Meta [have been doing this](#).

Expect to create immediate attrition risk by reducing people's compensation. Take this quote from an engineering manager working at a global tech company with a London HQ:

"Because of the pandemic and remote work, my company allowed for remote work, but changed how compensation bands are calculated. As I moved to another country in Europe, my salary was reduced by 30%, benchmarked to be 70% of what a person doing the same job as me is paid in the London HQ.

"As a manager, I understood the rationale. Still, it left a sour taste in my mouth and I set a goal to find a next-level job with compensation not calculated by location. I succeeded in doing so, and I'm starting as an Engineering Manager at a startup

paying location-independent. My new compensation is slightly above what I was making in London."

**Equity compensation drops** are a special case at companies where senior-or-above engineers and managers get a large part of their compensation as equity. At times when the value of equity goes up, this equity serves as an additional retainer. However, when it goes down, it results in less total compensation – effectively a pay cut.

In [The Scoop #6, I cover](#) unhappiness with compensation at many publicly traded tech companies, whose stock price has dropped around 50% or more, during the past 6-12 months.

If your company has seen major stock value drops, expect that plenty of people are considering leaving, unless there are measures that offset the accompanying fall in total compensation.

## Environment

A common reason people start searching for new jobs is dissatisfaction with their environment. Common reasons for dissatisfaction include:

**Psychological safety.** Do people experience discrimination or gaslighting? If so, they are unlikely to stay for long. Lack of psychological safety is often more common for minority groups in most workplaces.

As a manager, if you are not in a minority group yourself, you might not even notice issues of psychological safety or discrimination, being unaffected by them. Step up by actively looking for these, and become an ally to those who experience them.

**Diversity and inclusion.** How inclusive and diverse is the working environment? Are there senior tech leaders from minority backgrounds who serve as role models? Without a culture of diversity and inclusion, you will struggle to retain minority groups. Read more on [advice to hire and retain a diverse team](#).

**Work-life balance (WLB) and stress.** How long are the hours? Can people leave work behind when they wrap up, or go on holiday? Is overtime or working during weekends

common? How stressful is the job?

Everyone will have different views on how reasonable their work-life balance is and how justified the stress levels are. If people consider WLB to be poor or that stress is too high, they will leave. Get a sense of what people's needs are and compare with how they experience WLB and stress.

**A sense of belonging.** How connected do people feel to their teammates, their team and the company? How much do their manager and team mates care about them and how do they express this?

People who have little to no sense of belonging are more likely to view the position as "just another job" and are more likely to leave for a job which offers more money. They don't feel particularly attached to the team or company, so why not quit?

Full-remote work brings the risk of a lower sense of belonging versus regularly seeing colleagues in-person. This doesn't mean you cannot build up a similar sense of belonging in remote workplaces, it just means you need to be more deliberate about it and put in extra effort.

**Flexibility and red tape.** How easy is it to move hours around when something important comes up? How much admin does it take to get simple things done like requesting holidays, or purchasing tools and equipment that help people become more productive?

Lack of flexibility and too much red tape contributes to an environment people may find frustrating to operate in. An example of this is the amount of energy managers or engineers need to spend to be allowed to expense a \$500 noise-canceling headphone.



Gergely Orosz  
@GergelyOrosz

Engineer: "Can I expense this \$500 noise-canceling headphone? Makes it easier to focus."

Manager: "Sure."

Compliance: "That's against policy."

Manager: "It's \$500. 0.3% of their salary. It helps them do better work."

Compliance: "It's policy."

Manager: "Ok, how about this:"

10:17 AM • Sep 28, 2021

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**Friction in getting things done.** How easy is it to do day-to-day work? For engineers, how easy is it to run the tests, to do a code review, to push code to production? For managers, how easy is it to do the reports they are asked to complete by their manager chain, and how laborious are promotion or performance processes?

Poor tooling, legacy and fragile code, and lagging years – or decades – behind up-to-date practices, all contribute to friction in getting things done. The same goes for bloated management style and sluggish processes for managers. It's rare for people to point to these issues as a reason to leave, but they can contribute to a sense of working in a poor environment.

**High talent density.** What is talent density? [Louie Bacaj](#), former director of engineering at Walmart defines it like this:

"People want to work with other great people. If they tolerate low performers it will lead to attrition."

The way I think about talent density is the amount of outstanding people you work with. If someone works with a group of outstanding folks, but those people start to leave one by one and the "talent density" goes down, then this might make others wonder if they should also look elsewhere, or follow these highly talented folks to where they are going.

## Caring

How much does this person care about working on the team, and how much do others care about them? Some things to consider.

**Purpose of the team company.** As [Robert Lawson](#), software engineer at Auth0 puts it:

"When you've met most of the needs on Maslow's hierarchy, it's really compelling when someone can offer you a job with a purpose."

This works the other way around, as well. If your workplace doesn't offer much purpose, expect higher than usual attrition, unless you make up for it in one or more of the other areas. Counter-balancing with money is the most obvious solution; look at why small hedge funds – whose purpose is to make a few already wealthy individuals even wealthier – often pay the top of the market. If they would not, how could they motivate people to work there?

**Being heard and seen.** As a manager, do you ask about and learn people's needs? And do you do what you can to meet reasonable requests?

Companies which deny employees things that are cheap to grant – like flexible working, training opportunities, or expensing reasonably priced resources relevant for work – should not be surprised when the same employees feel undervalued. The same goes for things money can't buy – but which attention and time can – such as helping people achieve their goals, or giving feedback for growth.

**Acknowledging good work.** When team members get things done, make sure they get at the very least, a public thank you. Can you reward them in other ways? Awards, smaller grants, small gifts? All of these matter.

And don't forget to celebrate the wins. It's too common to move on to build the next thing after shipping a feature or product, without missing a beat. If you're a lead or manager, resist doing this before holding at least a small, fun celebration on what you, as a team, achieved.

**Caring about the work done.** Does the person care about what they work on? Sure, we all have work we don't particularly like or care about, every now and then. However, if someone is stuck doing work they dislike – such as a product-minded engineer

tweaking the build systems for months without an end in sight – those people will lose their own purpose, and can become a flight risk.

**Small gestures go a long way in making people feel you care.** This is especially true when the environment outside work is challenging. Small gestures might mean giving a day off for the whole team to recharge. I could be organizing a team event during working hours. Or offering to pay for movies, dinner and a babysitter for their family after a long project, when people worked overtime. Gestures show you care and help build personal connections.

## Growth

“Am I growing professionally in my current job?” This is a question everyone should ask themselves every few months. If the answer is “no”, then you’re getting stuck. And stalling in growth can be a reason for people to look for other opportunities where they can get back to growing. Factors which foster growth include:

**Achieving goals.** Everyone has goals. These might be team goals, project goals or personal ones. Someone achieving their goals will feel like they are growing. Those who don’t meet them will get frustrated.

The most common reason I’ve seen people consistently not achieve their goals is hectic workplaces with continuously changing priorities and frequent reorganizations. People set goals, but the goal posts are moved midway, making them feel they’re treading water.

**Professional growth opportunities.** Which options are on the table to challenge you? Are there impactful projects you can take on, or opportunities that can lead to career growth?

For ambitious people, the lack of growth opportunities can be a reason to consider either moving teams internally, or looking for external opportunities.

**The company’s growth trajectory** can also play a large role in how people perceive both their growth opportunities and their job security. If they have significant equity,

they will value that equity differently at a company that's growing, versus one that is shrinking.

While most companies keep growing, there are an increasing number of struggling companies, and several which are considering or executing layoffs. In a company that struggles to grow, some people will decide to look for other places with either more growth potential, or more stability.

**Mentorship and being invested in.** How supported is a person in growing professionally? Having one or more helpful mentors at the company can make a major difference. This is why better tech companies encourage [engineers to mentor other software engineers](#).

Apart from mentoring, how much are people being invested in? Are they getting additional support from management to take on challenging projects? For aspiring leaders, do they have support to join internal leadership programs, or external trainings, or conferences? Don't underestimate the difference it makes to some people to know that their management is explicitly invested in their growth. This is especially true for tech leads and engineering leaders.

**Teaching others.** For more experienced people, do they have opportunities to teach or mentor, and do they practice this? Teaching is a great way to not only develop less experienced colleagues, but to motivate experienced engineers and managers.

When I was at Uber, one of my favorite things to do was give a presentation at Ubersiversity – the onboarding program for new Uber hires – about how engineering works, and to take questions. This event was a crowd of 50-100 enthusiastic new hires who then bombarded me with questions and shared fascinating experiences. I walked out of these events with more energy than I went in with; and talking with other presenters, they had a similarly positive experience.

While not everyone will get as much of a thrill from public speaking as me, the act of helping others get better is very rewarding for most of us.

## Retaining early-career engineers

Early-career engineers have little professional experience and are usually working their first or second job. We're in a market where there seems to be far more entry-level engineers looking for jobs than there are positions available. So why should we talk about retaining this group? It's because this is a growing pain point.

### **Retaining early-career engineers is exceptionally challenging at many companies.**

I've talked with hiring managers at agencies and startups which hire more entry-level engineers than most, invest in them and help them grow. And yet, they see these engineers "head hunted" as soon as they've spent 6-12 months at their company.

Companies that do a great job growing entry-level engineers are often targeted by Big Tech firms, which take their early to mid-career colleagues. As a CTO at a digital agency shared:

"We have a reputation of being an agency with a very strong engineering culture. We are also good at hiring people with close to zero experience, and help them become great engineers.

The past few years, we've noticed recruiters from Amazon and other large tech companies targeting anyone who has been with us for at least 6 months. We cannot compete with these offers, and have been losing lots of great employees."

So, is there anything you can do to retain early-career colleagues, especially when you cannot compete financially with some players on the market? Turns out there is plenty.

### **Money & career**

- **Career path.** Many early-career engineers leave for other companies because they don't know a career path exists for them, beyond a months-long onboarding process. So create a multi-year career path with clear promotion points, and communicate what people can expect. A multi-year plan that the company delivers on, has been the single best retention tool for all agencies and startups I've talked with.
- **Promotions.** Be clear when these engineers can expect promotions and what they need to do for one. Early-career people are almost always ambitious and want to

get to that next level. If the next promotion level takes most people more than two years to reach, consider introducing interim levels to help with formal career progression.

## Growth

- **Make it clear what you invest in and how.** If engineers don't know what professional investment they are getting, they might not even realize how much you are helping them. Outline the resources they are getting; be it training, mentorship, events, stretch projects or others.
- **Mentorship.** Having regular access to experienced mentors is something most entry-level engineers value highly. However, few get this, and even fewer realize they are getting mentorship when it's called something else. Consider introducing a more formalized mentorship program to bring clarity for everyone: both the entry-level engineers who benefit and the mentors, so they know what support they are expected to provide.
- **Make them feel a valuable part of the team.** Involve entry-level engineers in all the initiatives you can. Have them be part of design discussions. Consider having them [shadow interviews](#). Onboard them to oncall – if they are up for it. Give them stretch assignments if it makes sense to do so. One of the best ways to help these engineers grow is to make them full members of the team, and support them until they have the skills that most of their colleagues do.

## Retaining software engineers

It's never been harder to retain software engineers with experience. The job market is hot, and we might be at the start of The Great Reshuffle.

### Compensation

Be aware of how competitive the market is, and how much it has moved up. Know that replacing an experienced engineer will likely cost a lot more than what their current package is, so consider investing not just in hiring, but in retention.

### Environment

- **Oncall.** People don't just leave managers, they leave terrible oncall rotations as well. Make sure yours is not terrible.
- **Friction with development that adds up.** DX founder Abi Noda shares:

"Tooling issues, process problems, or suboptimal team dynamics... These can all lower engagement and lead to attrition. A familiar example might be not having enough uninterrupted time to do deep work. That's painful — developers don't feel like they can have the impact they'd like to have. If unresolved, it can build up and make a new job look more appealing."

- **Tech debt that makes development frustrating.** Working with legacy codebases that have no safety nets like automated tests, or having to implement a feature twice – once for the new, and once for the legacy system – are examples of tech debt which adds more friction to development work. Figure out what the truly frustrating parts are, why they exist, and aim to pay down those parts.
- **Roadmap visibility.** What work is coming up in the next 3-6 months? CTO Sergio Pereira suggests visibility of the long-term helps people trust they are in a stable environment:



Sergio Pereira 🚀  
@SergioRocks

[@GergelyOrosz](#) Most of my reports get really engaged when I give them details into 3-6 month roadmap.

I tell them the initiatives they'll be involved with, and discuss openly their status, even if those initiatives are still far from development ready.

People like to know what's coming up.

5:32 PM · Mar 24, 2022

## Caring

Ask about colleagues' career and professional aspirations, not just within the company, but what their goals might be beyond it. As engineers become more experienced, their

priorities shift and goals might range from moving up the existing career ladder, through to starting a company soon, all the way to saving enough money to open a bakery in a few years' time.

Get to know what people care about and then you can help where you can with these goals.

## Growth

- **Opportunities to lead and to grow to the next level.** For engineers at or above the senior level, some will want to move into positions with more leadership. Do they have the possibility to do so? If not, is this advancement important enough that they might leave the company for it? Can you help them move internally, to a team with such opportunities?
- **Firefighting.** Is there too much dealing with crises going on? If so, on top of possibly burning people out, firefighting will get in the way of working on higher leverage work that can get people to the next level.
- **Time and opportunity to teach and mentor.** It's hard to grow above a certain level without teaching or mentoring others. As a manager, help create these opportunities and the dedicated time needed for mentoring or teaching others, including preparation for these sessions.

## Retaining engineering managers

While most companies go to great lengths to retain software engineers, I've noticed that managers of the same engineers don't get equal attention. Directors-and-above often assume that managers will stay and so take no preventative steps against attrition of managers.

My experience shows that while managers do tend to change jobs less frequently than most engineers, it pays to not take this for granted. Backfilling an effective engineering manager is much harder and takes longer than replacing an engineer.

Here are things I recommend to retain managers:

## Money & career

- **Understand their current career goals.** Many engineering managers took a step down during the pandemic and full-remote work. I've talked with several EMs who are now looking to take on a bigger role, and yet their leadership has no idea of these aspirations, as they do not ask. It's only the recruiters they are talking with who understand what they want. Have you asked your managers how their career aspirations have changed?
- **Detect and act on compensation friction.** Most managers are tactful enough to not bring up compensation issues as directly as engineers would. But this does not mean issues don't exist. For example, managers are aware that offers for new hires are much higher than before, sometimes even above their own compensation. As a manager of managers, don't let compensation friction simmer for too long: address these issues head-on. Have a frank conversation on what you want to do, what you can do, and what you will do to resolve it.

## Environment

- **Fight for budget for the manager and their team.** The market is moving up. Push to get budget to both hire engineers and managers, and to retain them. If you cannot retain engineers and struggle to backfill them, it's only a matter of time before your manager also leaves.
- **Keep an eye on hiring.** If your EMs struggle to hire, they will become a flight risk. Who wants to stay at a company where they cannot grow their team, either due to an overly low budget, overly constrained working environment, or an uninspiring mission?
- **Get rid of red tape.** What unnecessary administrative processes are tying up managers, and causing pointless frustration? Use your authority to remove these.

## Caring

- **Sit down and listen to them.** When was the last time you did this? I mean, when you truly listened. I don't mean giving updates on the organization, or sharing the next big thing in the pipeline. I mean asking them what's on their mind, and having a conversation about this.

- **Challenges they have.** What are the most pressing ones and why? Can you work through these with them?
- **Take something off their plate.** If they have too many things on the go at once, can you empower them to say “no” to some work, or even take something off of their plate by allocating it to another team, or you taking it on?
- **Reduce expectations and have their back.** Is there attrition on their team? Is one of their projects delayed? Support them in doing the right things: pushing back on other projects, turning down important but not top priority work, and other ways you can help. Let them know they have your support, and deliver on that promise.

## Growth

- **Understand their growth goals.** As mentioned above, many EMs might have adopted smaller roles during the pandemic and now may want to take on larger projects and teams. Some will have the opposite goal; to have smaller teams while they focus on external projects like moving house. Understand what they are looking for and create space for it. Expect they will leave if you don't know what their goals are, or don't support them.
- **Invest in your managers.** Most managers of managers forget that EMs appreciate investment just as much – if not more – than individual contributors. Investing can mean recruiting them to larger initiatives. Set them up with mentors they look up to. Get manager coaching for them. Enroll them in training programs they are excited about or leadership conferences they want to attend.
- **Give them ownership of something they care about.** There's usually a lot of things to fix. Can they take the lead in one of the areas that is a pressure point for their team?

## Takeaways

When I was a manager at Uber, I expected people to leave my team and the company, sooner or later. However, instead of just accepting this was how things were, I did what I could to create a great working environment in which people felt cared for and had growth and career opportunities.

The result? In four years, out of about 30 engineers in my organization, one left the company, and four transferred to other teams, with all transfers working out as successful career moves for them. I cannot take credit for things beyond my control, such as the compensation budgets or the aspect of luck.

Since then, I've observed certain managers, directors and executives become a lot better at retaining people. And most of the ways they made this change are summarized in this article.

Figure out what is important to people. Take stock of how people feel about these four things:

- Money and career
- Environment
- Caring
- Growth

Make sure that in these areas, they are not unhappy with too much. If you do this, then you'll have fewer problems with attrition, even in this historically red-hot hiring market.

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